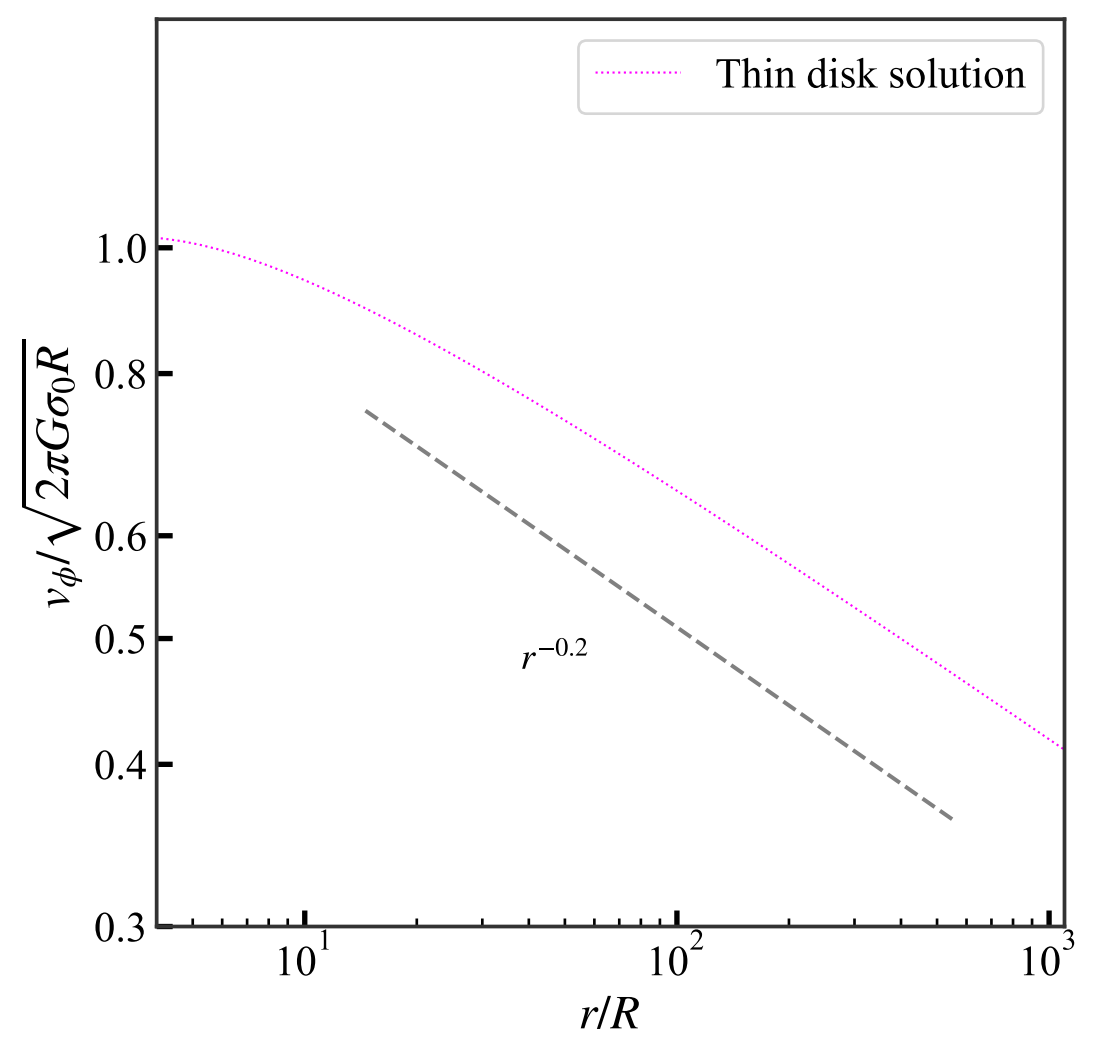
Thin disk solution
$r^{-0.2}$
$v_\phi/\sqrt{2\pi G\sigma_0 R}$
r/R
1.0
0.8
0.6
0.5
0.4
0.3
10^1
10^2
10^3